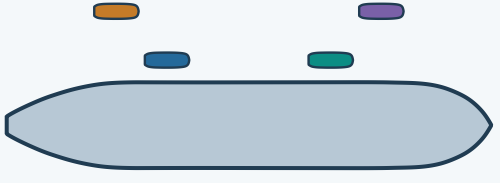
Coupled ship–tug dynamics



Two push + two tow

Independent local NMPC

Tug 1

Tug 2

Tug 3

Tug 4

Shared ship state · no peer commands

Directional evaluation

16 / 16

tested cases reached the terminal envelope

37.6–96.1 min